

Skills and Knowledge in Cloud Learning Platform

In Cloud Learning Platform one of our goals is to enable the management of skills and knowledge. Another goal is the effective delivery of those skills and knowledge to learners at scale. In this document we explain the strategy for skills and knowledge in Cloud Learning Platform, including:

- Definitions and distinctions between skills and knowledge
- How we represent skills and knowledge in data
- How we represent skills and knowledge in the form of “learning graphs”
- How skills and knowledge enable customer use cases
- Why we represent skills and knowledge in separate data structures
- What capabilities for skills and knowledge management are included in the platform

The difference between doing and knowing

We start with defining what we mean by skills and knowledge. These are definitions that we hope readers will find reasonable, but they are nonetheless definitions. They are the axioms we will assume, if you will, for our treatment of skills and knowledge. Our framework borrows from computer science and cognitive science, but it should make sense to those who do not have backgrounds in those fields. We all possess skills and knowledge, and brains, and so we all have an intuitive sense of what those things are and how they work.

A skill is something we can do. It is like a cognitive “program,” similar to a computer program, but using the computational machinery of our brain, which we assert is not the same kind of machine as a symbolic computer. We won’t delve into that difference here; the point is to make an analogy between skills and programs that are “cognitive,” executed on the computational machine of our brain.

A cognitive program takes inputs, just like a symbolic computer program, and produces outputs.¹ A simple example: “recall” is the skill of retrieving knowledge from memory. The

¹ For those familiar with theoretical computer science: our intention here is to define cognitive skills as recursive functions, just like computer programs. However a cognitive skill as defined here is a mathematical generalization of a computer program. It is a recursive function that executes continuously until if and when it halts, mapping a set of inputs to a set of outputs, but it is fundamentally stochastic in its behavior in a way that Turing, natural number-based computation is decidedly not. And cognitive skills operate over a different input domain than Turing computation, which is based on the natural numbers. For those familiar with the math of differential geometry: cognitive skills can take perceptual input which may be represented mathematically on smooth manifolds. Cognitive skills also take as input the output of other skills. The output space of skills may also be represented on smooth manifolds. This framework provides a connection between skills and machine learning and neural networks, which also may be defined to operate on smooth manifolds in $\mathbb{R}^n$.

inputs to the recall skill are a “query” for our memory: a mental representation that often contains part of the knowledge we are trying to retrieve (“what was the name of the singer who sang that song?”). The output of the “recall skill” is some knowledge from our memory. The output of this skill often serves naturally as input to other skills.

Recall is a “mental skill” in that it doesn’t produce physical action. There are many other kinds of mental skills, such as reasoning and inference, problem solving, and decision making. And of course there are physical skills. Physical skills are also executed on the computational machinery of our brains, and produce output which are control signals to our bodies.

Knowledge is something we “know”, and so as not to be circular in our logic, we define “knowing” as making assertions about the state of the world, or properties of some entity or category of entities. By entities we mean “things” in the universe, either real or abstract. To extend the analogy with symbolic computation: knowledge is data that represents the world. Skills are programs that we use to act on the world.

To provide a more concrete example of the difference between skills and knowledge, consider the job of a medical practitioner. To do their work they must know about many things, like anatomy and physiology, disorders and diseases and their associated symptoms, and medications. They can demonstrate their medical knowledge by reciting on demand a list of diseases and their symptoms, a list of medications used to treat those diseases, and for each medication a recommended dosage, and list of side effects.² But we can all intuitively appreciate that having lots of knowledge does not by itself mean we can do anything useful with that knowledge. Just memorizing a list of diseases and their symptoms and drugs used to treat those diseases does not mean that one has the skill of treating any particular disease.

Skills make use of knowledge. The knowledge a medical practitioner possesses enables them to do useful things to treat patients. The skill of diagnosing a disease requires knowing about the respective symptoms and how they manifest physically, as well as the underlying physiology affected by the disease, as well as causal relationships between the physiology and the symptoms. But performing the skill of diagnosis involves more than just recalling all those facts: one must execute a series of steps or actions that rely on that knowledge, including observing the patient, asking the patient to report their symptoms, making measurements of physical attributes using tools like heart monitors and blood pressure cuffs and blood tests, and weighing all the observed factors to determine the most likely underlying disease that caused them, based on the practitioner's knowledge of diseases and symptoms.

² The act of demonstrating that I possess that knowledge employs a skill: the skill of recall. Remembering things I have learned and reciting them is a skill that we can learn and improve.

The output of the “diagnosing a disease” skill is a diagnosis. Let’s consider how we evaluate the quality of that output. The accuracy of a diagnosis, with the according implications for the practitioner’s proficiency in the skill of diagnosing disease, may not be determined until the patient has a response to the treatment. This raises a few general points: a) in order to know the proficiency of a skill there must be some general process to evaluate the quality of the output, b) in this case, the output of the diagnosis skill must be supplied as input context of the other skills, namely the skill of determining a treatment for a disease and the skill of applying a treatment. So skills can chain to other skills, adding their outputs to the context of the next skill, c) we can’t evaluate the output of the diagnosis skill until we have performed the complete skill chain of determining treatment and applying treatment, and d) the entire chain may be viewed as the skill of “treating a patient”, thus e) skills are composed of subtasks which may be viewed as skills themselves, so there is a fractal quality to skills. Skills are composed of skills which are composed of skills.

Representing skills in data

In Cloud Learning Platform we use this conception of a skill:

A skill is an action performed by a learner that can be repeated consistently, that has an initial context and end goals.

The initial context for the skill is the set of conditions which must exist in order for the skill to be applied. This context forms the input to the skill. The action of a skill consists of a graph³ of steps or “subskills,” that are executed in a sequence. The end goals for the skill allow us to know when we are finished with the skill, and define a quality metric for the output of the skill so we can know whether we have achieved the desired outcome. Our proficiency in a skill is defined by our ability to apply the skill in the appropriate context, to repeat that skill reliably, and to consistently achieve the end goal or goals of the skill.

For the skill of throwing a baseball, the initial context is both physical (the ball is held in the hand of the arm which will be used to throw, there exists free space around the body to allow the arm to move) and mental (the thrower has a notion of the direction or target of the throw). The action sequence includes moving the arm backwards, stiffening the wrist, and once the hand and the arm has reached a certain position then propelling the arm forward. The end goal is that the ball is propelled through the air towards the target. For a skill like solving a particular kind of math problem, the initial context is largely mental (the solver has correctly interpreted the mathematical equations and language description of the problem) though there will always be some physical context as well (the problem is described on a sheet of paper in front of the solver, the solver is holding a pencil and knows the place where they must write the answer).

³ this graph is like a “flow chart” - see more detail below.

The action sequence will include mental steps (performing internal symbolic manipulation of the equations in the initial context) and possibly physical ones (writing down the output of the intermediate symbolic manipulations). The goal might be a written answer that is “correct” according to the particular logic of that class of problem, possibly along with a written sequence of intermediate steps that “show the work.”

To create technology to manage skills and assist the teaching of skills, we need to represent those skills as data somehow. Each component of skills, the context, the task sequence, and the end goals, comes with unique challenges in data representation. Creating data models for the context in which a skill is applied could require creating models of the physical state of the real world, which is hard. It is also challenging to fully represent a sequence of actions performed by a human which involves a combination of physical and mental tasks and decision points between those tasks. We can represent such a sequence using something like a “flow chart,” where each step in the flow is a particular task, and decision points allow for proceeding down a different path in the flow. However each task itself can always be decomposed into further “subtasks” which have their own flow and decision points. Thus representing the action of a skill means having a generic representation of branching operations that unfold over time. Creating data models for goals is challenging in a similar way to representing context, since knowing whether the goals were met means knowing about the state of the world, which is complex to represent and evaluate. To represent skills in data we clearly must adopt simplifications in how we represent each of the components of a skill.

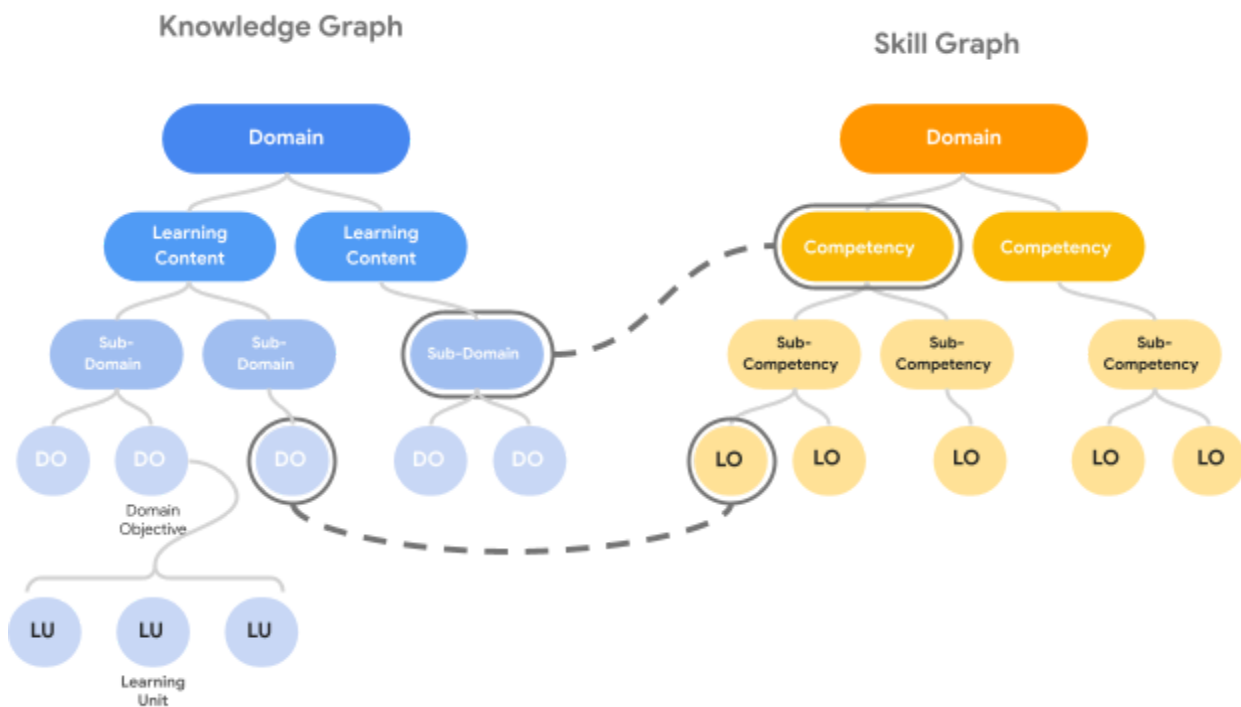
In practice in education we typically have one data object to represent a skill, which can be represented as a “node” in a graph (which we will describe in more detail below in the section on learning graphs). We use different strategies to represent the context, action sequence and goals in data. We can represent the context for the skill via relations between that skill and knowledge. For the subtasks we can rely on compositional relations between skills and subskills. To know whether the goals of the skill have been met we rely on assessment associated with the skill, which may involve the evaluation of the output of the skill against a rubric created by humans. Let’s examine each of these strategies in more detail.

Representing context

In the real world the context for a skill is much richer than just a list of things that must be known prior to performing the skill. To perform a skill we must have a “world model” for the situation in which the skill applies. For example, to pay my bill at a restaurant, I must know who my server is, what the social conventions are to request the bill from the server, how to review the bill for accuracy which requires knowing what everyone at the table ordered, what my options for payment are, whether or not tipping is required according to social conventions and how much to tip based on the bill total. These are all part of a “world model” of the scenario of being in a restaurant, ordering and paying for food. In practice we cannot hope to capture

anything close to such a world model in our data. But by mapping skills to knowledge, essentially making a link between each skill and the knowledge required to apply that skill, we can capture some of this necessary context. Essentially this mapping encodes the inputs to a skill: the things we must know in order to execute the skill successfully. This is one example of the usefulness of differentiating between skills and knowledge.

Adopting this scheme for representing the context of a skill also allows us to objectively define interesting metrics for skills, such as “abstractness” or “generality” and even “usefulness.” Skills that are mapped to large sets of knowledge as their initial context are by definition applicable in a broad set of contexts. Skills that we care about in education like “problem solving” or “constructing an effective argument” fall into this category.



Mapping between skill and knowledge graphs

Representing action

To represent the sequence of actions that comprise a skill, we can represent the subskills for that skill. In general we can represent the relation between a skill and other skills through edges in a graph, as we will explain later. These relations can include “compositionality”, meaning we can represent that a particular skill is composed of other skills, or subskills. You can always decompose a skill further into subskills, until you get to skills that reference perceptual tasks or cognitive abilities or physical actions which we lack a good framework to

represent in data. In this scheme the notion of “atomic skills” is useful: this is a skill that is independently measurable from other skills, such that our ability to perform that skill does not provide additional information on our ability to perform any other skill. If we wish to be exhaustive, we can recursively decompose skills into subskills until the subskills are atomic.

We can think of the action of a skill as a “cognitive program.” Both mental skills and physical skills are represented and performed by our brains using similar cognitive machinery: the cortex and the cerebellum. The physiological structure in these parts of the brain are uniform across the brain, suggesting the brain machinery associated with physical skills is the same as those for mental skills. This also suggests that there is a unified underlying mental representation of the action for both physical and mental skills. We do not have an accepted neuroscientific or computational theory for that unified representation, but we can posit some mathematical abstractions that are general enough to encompass whatever form that theory might take. We know that a program requires input data. The context for the skill is like the input to the cognitive program. We know that the action of skill unfolds over time. So we can represent a skill as a mathematical function, with the input context and time as parameters, and the output of the function as the evolving state of the cognitive machinery of the brain. This kind of [dynamical system](#) is an established approach for describing mental processes in cognitive science. So a skill can be represented mathematically as a dynamic function (with time as a parameter), with the input context being the domain of the function. The size of the domain for a skill function is a measure of the abstractness or generality of the skill. Learning a skill means acquiring a skill function with our cognitive machinery.

Representing goals

In education representing the goals of a skill is achieved implicitly, through associating assessment with the skill. We use assessment in education to evaluate whether a learner has achieved proficiency in a skill. Proficiency means they have acquired knowledge of the context of the skill, have learned the action sequence and can perform it reliably and effectively in the appropriate contexts, and they achieve the goals of the skill consistently across those contexts. Assessment is a deep topic in itself which we won’t cover here. But it is important to note that to complete the data model for skill by representing the goals, we must associate an assessment to that skill.

Returning to our medical practitioner example, the goal of the skill of diagnosing a particular disease is to correctly select that disease from all the possible diseases, based on the symptomology of the patient. We can represent the goal of that skill by associating an assessment to that skill which could consist of a number of different scenarios, consisting of sets of symptoms manifested by a patient. The learner is required to select which sets of symptoms are most likely to correspond to a patient that has the disease.

Knowledge representations

In Cloud Learning Platform our approach to representing knowledge follows the conventions of [semantic ontologies](#). This approach is very common in the fields of data science and artificial intelligence. Knowledge is defined as data that represents a set of “entities,” and relations between those entities. By “entities” we mean things, including physical objects, people, places, organizations, as well as “concepts.” The notion of a concept includes categories of entities that are not physical objects. Abstract concepts like love and loyalty, or concepts in domains of knowledge of science or mathematics, such as entropy, information, computation and manifolds, are things we can say we know something about, and refer to as entities, but clearly do not correspond to physical things.

Relations between entities can include anything we could denote with a verb or predicate phrase, like “causes”, “contains”, “is the parent of”, “is the child of.” The latter “compositional” relations are of particular importance since they allow us to define a hierarchy of entities and concepts, in which entities are composed of other entities, or “subentities.” For a physical object the subentities could correspond to the parts of the object. So a car would have “is a parent” of relations to its parts, including wheels, doors, an engine, etc. For concepts, the subentities could correspond to “sub-concepts.” The mathematical concept of derivative might include the sub-concepts of slope, tangent, limit, and infinitesimal.

As a data structure, this kind of hierarchy is naturally expressed as a tree graph, which we go into more detail below. Both physical entities and abstract concepts can be expressed in semantic ontologies as an entity with an associated network of subentities with defined relationships to each other.

Entities also have attributes - properties or features of the entity that distinguish it from other entities. So the complete data representation of any entity includes the attributes of the entity in combination with the relations that of that entity and its subentities.

Academic knowledge

In recent centuries as human civilization has developed systems of education, knowledge has been organized into categories corresponding to academic disciplines. Each field has its own conventional taxonomy for organizing knowledge into subdomains and topics. The conventions at work here are both social and evolutionary, defined by the collective consensus of experts in the domain that evolves over time. The easy way to get an idea of the current consensus for the knowledge taxonomy of an academic field is to survey the introductory college textbooks for that field. The units and chapters of those books together define a consensus of the hierarchy of topics and subtopics. Some academic fields are more rigorous than others about defining the taxonomy. There are many fields in which there is a commonly

referenced book entitled “Handbook of X” that is blessed by a professional organization associated with the field, which provides in a sense an “approved taxonomy” of knowledge for that field.

What will become apparent to anyone who goes through the exercise of comparing widely adopted introductory college texts for any field is that the consensus is not a perfect one. Each expert author of a Psychology 101 textbook will have a different idea of how to organize topics and subtopics into units and chapters. The process of teaching is itself a skill, of transmitting skills and knowledge in an action sequence. Writing textbooks, or in general crafting teaching materials, is also a skill. The context of the skill of writing a textbook for a field is an arrangement of the knowledge of the field. A textbook author starts with their own mental model of the state of knowledge in the field, and adds to that knowledge the accumulated research that has taken place in the field since it was founded. Each teacher or textbook author arranges their internal knowledge model to suit their preferred method of telling the story of the topic they wish to convey.

Just as each expert will have their own arrangement of knowledge and skills in their internal knowledge base, schools will also have their own skill and knowledge taxonomy. In some areas of education external taxonomies are more strongly adopted. In particular primary education is more likely to adopt such standards given the scale of that form of education (every child must go to school), and the societal desire to enforce high standards of academic achievement for developing members of that society. But in general, schools must know what skills and knowledge they wish to convey, and how to arrange those skills and knowledge in offerings to students.

Skills as knowledge

Skills are a special form of knowledge. Skills are entities with attributes, subentities, and relations to other entities of the same category (other skills). These can include precedence relations (you must learn skill A before B) as well as compositional relations (skill A may be decomposed into subskills B, C, D). It is a common pedagogical approach to teach a skill by first teaching the subskills, then teaching them together to teach the “parent skill.”

The sources and repositories for the knowledge of skills are distinct from general knowledge sources. You can learn the context for a skill from sources of knowledge like textbooks, but in order to learn the action sequence, to acquire the cognitive skill function, it is a foundational pedagogical principle that one must practice. One must repeatedly attempt to execute the cognitive program of a skill in a proper context and evaluate whether the goals are met. One must learn how to do by doing. Whereas non-skill knowledge can be learned by rote memorization, or via direct experience, or via language.

Language is a sense for knowledge, in a sense. Language is the skill of “serializing”, or “deserializing”, knowledge. In software engineering we call the process of writing out a data object as a sequence of bytes “serializing” that object. Language is the process of serializing world models, our internal data model for representing the state of the world and our intentions, which can be described as entities and relations between entities. We serialize our world models into words, and we deserialize words into world models. We can represent world models, a collection of entities and relation objects, as graphs, to which we turn now.

Learning graphs

The data structures for both skills and knowledge take the form of a graph: a tree in which the top level nodes represent more abstract knowledge components or skills, and the lower level nodes represent more granular, detailed knowledge and skills. We refer to those trees as “learning graphs.”

A learning graph tree may be naturally organized into “levels,” where nodes are grouped into categories according to their distance from the top of the graph. The names used for these levels are typically going to differ widely between different educational institutions. In Cloud Learning Platform we allow the names of the levels of knowledge or skill graphs to be customized, allowing customers to use their existing naming conventions if they desire. The recent trend in “competency-based” learning means that the top level of a skill graph are often labeled “competencies.” A competency is a skill that encompasses many skills associated with an academic discipline or profession. There might only be 20 to 30 competencies associated with a bachelors degree. Each full credit course in that degree program might be mapped to a few (say 2-5) competencies.

A learning graph for skills mirrors the popular Blooms taxonomy. Higher level skills like problem solving or “constructing and defending arguments” would show up in the higher levels of the skill tree. Lower level skills like organizing, summarizing or translating would show up in the lower levels of a skill tree. The leaf nodes of the tree, the skills at the very bottom, correspond to simple “recall” of items of knowledge. If we are mapping skills to knowledge this would imply that each entity in a knowledge graph could be associated with a recall skill, as a leaf node in the skill graph, for that knowledge entity.

We can also represent that skill A must be learned before acquiring skill B through precedent relations in the graph, using directed edges.

The learning graphs for academic knowledge are essentially just knowledge graphs with a predefined level hierarchy that is aligned to the domains of knowledge organized according to academic tradition. The academic domains of knowledge include domains for each of the major fields of human knowledge commonly taught in schools: mathematics, physics, biology,

chemistry, psychology, history, art, etc., divided into subfields within those domains (newtonian mechanics, quantum mechanics, microbiology, neuroscience, materials science, organic chemistry, etc).

Linguists and philosophers of knowledge are often concerned with how we organize knowledge entities into categories. A category, or class, of entities are all related to each other through possessing similar attributes, or similar relations to other classes of entities. One useful way to think about this is geometric: if entities of knowledge may be considered to exist in some abstract space, then entities that are similar to each other in some systematic way would be near to each other in that space.

For academic knowledge graphs the notion of “salience” is relevant. Some entities are more important than others to transmit to learners. Salience can be derived from measures of frequency (how often a concept or entity appears in textbooks) and relations (how many other entities does this entity relate to, and the nature of those relations). This is a slippery and even controversial topic. Salience is a measure of the value of knowledge, and just because a particular entity or category of entities appears more often in our knowledge graphs, or has over the history of human endeavors acquired more relations with other entities may not mean that we should spend more effort teaching those entities to students.

Learning goals

Every learner has a goal, implicit or explicit. Learning goals can consist of careers, missions, academic credentials like degrees or certificates, or individual skills. To help a learner achieve their learning goals, a school must know the skills and knowledge associated with those goals. The goal of Cloud Learning Platform is to help schools build their curriculum and learning experiences on a scaffold of skills and knowledge, where every academic offering is constructed from the ground up by referencing the skills and knowledge necessary for those offerings, and every learning experience such as an individual course is designed with the skills and knowledge woven into all the materials and assessment.

This approach will make it possible for schools to schools to provide learners:

- recommendations for learning goals that they may not have been aware of, that are similar to their stated learning goals
- a personalized pathway to achieving their learning goal, that is informed by the learners current skills and knowledge, and can be adjusted as life circumstances change or new opportunities arise.

Use cases for skills and knowledge

Use cases for institutions

How do learning institutions make use of skills and knowledge? When we refer to “learning institutions” we of course mean schools, in primary grades or higher education. We include any other organization that engages in large scale education or training such as corporations and public entities like state and local governments and the military. It is not controversial to say that the purpose of an educational institution of any kind or at any level, is to transmit skills and knowledge to learners.

In order to achieve that purpose the learning institution must achieve the following goals:

- **know** what skills and knowledge they wish to transmit
- **package** the skills and knowledge into offerings for learners, including learning experiences like courses, and academic products like certificates, credentials and degrees
- **manage** the skills and knowledge and the respective offerings built around them as they change and evolve to meet the needs of learners and the job market
- **link** skills and knowledge used in curriculum to **external standards and repositories** to assist in accreditation, ensure alignment with employer needs, and design effective academic programs
- be able to **assess** and **measure the mastery** of skills and knowledge in a variety of contexts, including to understand prior learning for a given learner
- understand **dependencies and prerequisites** for acquiring skills and knowledge, and instances when individual bits of knowledge or skills are re-used, repurposed, or leveraged differently while in other contexts
- efficiently **develop** and **maintain** teaching materials or “learning content” for skills and knowledge
- employ effective and scalable **pedagogy** for transmission of skills and knowledge
- be able to **measure the efficacy** of learning content, instructors, academic offerings in teaching skills and knowledge
- be prepared to **advise** students to help them achieve success in learning goals and learning paths

In the next sections we enumerate specific use cases for both educators and learners that highlight the use of skills and knowledge, organized according to those goals of the learning institution.

Note to avoid duplication in these use cases, we only mention a particular use case once, even though it may apply to multiple goals.

Educator personae

The following personae will be referred to in the educator use cases. Each use case will name the associated persona(e).

Academic Provost (AP): an executive level decision maker at a school in charge of managing existing academic programs and developing new ones.

Curriculum Manager (CM): a member of faculty responsible for designing curriculum (courses, syllabi, assessment, learning content) to support an academic program.

Instructional Designer (ID): a member of faculty (who may also be an instructor) responsible for designing and authoring learning materials (media like textbooks/images/video) as well as assessment.

Registrar (R): a member of the office responsible for administering student enrollments and credit transfers from prior academic experience.

Student Advisor (SA): a member of faculty who advises or counsels students on academic or personal issues. This role may also be partly played by

- 1) automated recommendation services for learning goals, learning paths, and learning activities
- 2) student advising chatbots.

Educator use cases

I. **Know what skills and knowledge the school wishes to transmit (AP / CM / ID)**

- a) When designing academic programs, academic provosts must know the set of skills and knowledge that are important to achieve the goals of the program (for example, to prepare learners for careers in a particular industry).
- b) When designing curriculum, curriculum managers must identify and organize sets of skills and knowledge that are intended to be transmitted to learners in that curriculum.
- c) When designing and authoring learning content, instructional designers must identify and organize the skills and knowledge intended to be transmitted by the learning content.

II. **Package the skills and knowledge into offerings for learners (AP / CM)**

- a) When designing academic offerings like degrees or credentials, curriculum managers must know the set of skills and knowledge that are important to achieve the goals of

prospective students (for example, to qualify learners for certain jobs in a particular industry).

III. Manage skills and knowledge as they change and evolve to meet the needs of learners and the job market (AP / CM)

- a) To design effective academic programs, academic provosts must know what skills and knowledge are required by employers.
- b) To design and maintain effective and successful academic programs, academic provosts need to know how skills and knowledge required by the job market evolve, including new skills that are required or existing skills that are less relevant.

IV. Link skills and knowledge in curriculum to external standards and repositories (AP/CM)

- a) To support accreditation an institution must be able to demonstrate the skills and knowledge associated with curriculum.
- b) To ensure students are prepared for jobs and careers an institution must align the skills and knowledge used in curriculum design to those that have been identified as important to careers and industries, as represented in external repositories.

V. Assess and measure mastery of skills and knowledge in a variety of contexts (CM / ID / R / SA)

- a) When designing curriculum, curriculum managers and instructional designers must include assessment items that test all the skills and knowledge associated with that curriculum. These assessment items should not test skills and knowledge that are not part of that curriculum.
- b) To award credit for prior learning or experience, a registrar needs to assess the skills and knowledge possessed by an incoming student.
- c) To design a personalized path for a student, an academic advisor must understand the level of mastery of skills and knowledge a student has at the beginning of their learning path, acquired via prior academic learning experiences or work experience.
- d) To know when a learner might need additional support, an academic advisor must be able to track the mastery of skills and knowledge during their learning journey,

VI. Understand dependencies and prerequisites for acquiring skills and knowledge (AP, CM, ID, SA)

- a) To design effective and coherent curriculum, curriculum designers must easily be able to track instances when individual bits of knowledge or skills are used, repurposed, or leveraged differently in all contexts (programs courses, learning content, assessment)
- b) To know whether a student is prepared for their learning journey, instructors and academic advisors must be able to provide prior learning assessment to a student, which requires knowing the dependencies and prerequisites for that journey.

VII. Efficiently develop and maintain teaching materials (ID)

- a) To efficiently retrieve individual content items like video, text and images associated with skills and knowledge, instructional designers need to categorize and organize those content items according to the skills and knowledge they convey.
- b) When authoring learning content, an instructional designer needs to be able to retrieve content items using queries that include skill or knowledge entities and relations.
- c) To support the automated generation of formative assessment using AI (Artificial Intelligence)

VIII. Employ effective and scalable pedagogy for transmission of skills and knowledge (ID)

- a) When designing learning materials, instructional designers know what learning outcomes the materials wish to achieve.
- b) When designing learning materials, instructional designers must employ pedagogical techniques appropriate to the learning outcomes they wish to achieve.

IX. Measure the efficacy of learning content, instructors, academic offerings in teaching skills and knowledge (AP, CM)

- a) When managing academic programs, academic provosts must understand the efficacy of existing curriculum and learning experiences (e.g. courses) in fostering student success.
- b) When managing learning experiences (e.g. courses), curriculum managers must understand the efficacy of these learning experiences in fostering student success.

X. Advise students on their learning journey (AP)

- a) When managing academic programs, academic provosts must make support available to students for learning competencies and domains of knowledge.

Learner use cases

As a learner I want:

- to know and have agency over my learning goals
- to know the skills and knowledge associated with my learning goal
- to have access to learning content for the skills and knowledge associated with my learning goal
- to understand prerequisite knowledge and skills for certain learning experiences, and for particular learning goals (such as jobs and careers)
- to have access to learning experiences associated with my learning goal
 - including practice opportunities to learn skills
- to receive feedback on my learning progress that is mapped the skills and knowledge I am required to learn
- to know when I have achieved mastery of a skill according to the standard of an expert

In the [Reference](#) section we provide some documents that discuss the roles of skills and knowledge in the management of a university, as well as specific use cases that have been identified by educators in that university.

Why represent skills and knowledge in separate graphs?

1) Knowledge and skills use different taxonomies and schemas, and come from different data sources

Knowledge and skills have different taxonomies that originate from the different data sources and use cases for these two categories of data. Even though skills are a form of knowledge, as was noted above, knowledge entities and skill entities represent quite different data items, and thus should be represented using different data schemas. The strategy of storing knowledge separate from skills allows us to create different schemas and taxonomies for knowledge and skills.

The levels of a knowledge graph may typically be structured into “domains”, “topics”, “subtopics”, and “concepts”. Knowledge entities derive from learning content sources like textbooks. In our schema for knowledge we include the notion of a “Learning Unit,” that represents a leaf node in the knowledge tree. These Learning Units are special objects (inheriting from the parent class of Knowledge node) that represent a “fact” that we wish the student to know. These “Learning Unit” facts consist of a central entity, which represents a “salient” concept or entity. For each of these salient nodes in the graph we associate a detailed knowledge graph that provides a richer knowledge context, as well as text samples from learning content that can provide a language context for the concept or entity.

The levels of a skill graph are typically labeled as “competencies”, “sub competencies”, “learning objectives” etc. Skill entities link back to the sources for skills that ultimately come from job related requirements, as well as (hopefully) an understanding of what core skills are needed to be successful in modern life. These are called “21st century skills” in the modern education science parlance, and include skills like “persuasion”, “formulating and communicating a strategy”, “teamwork, leadership and being an effective follower”, “framing a problem and proposing a solution.”

2) Storing knowledge separate from skills allows us to represent the context for a skill as a mapping between skills and knowledge

The mapping between skills and knowledge provides a richer representation of a skill, allowing us to define metrics like the abstractness and generality of a skill, the applicability of a skill in a particular domain of knowledge, and the difference between two skills based on their knowledge context. Computing the alignment between two skills, or effectively whether “skill A = skill B”, can be informed by knowing their respective knowledge contexts.

3) Users want to search and discover from both the skill and knowledge perspectives

Users want to query learning content and assessment based on both concepts and entities (“find all the assessment items related to diabetes”, “find videos on the topic of the discovery of insulin”) as well as skills (“find textbook chapters that explain protocols for treating chronic diseases”). Users want to query skills based on the knowledge they apply to, and query knowledge based on the skills that use that knowledge. A query like “Find learning content on medicines used to treat chronic diseases” mixes knowledge components with skills.

4) Pedagogical techniques for teaching skills and knowledge are different

Representing skills and knowledge in different trees allows us to respect the different methodologies used to teach skills and knowledge. Teaching knowledge relies on memorization and recall in context. Teaching skills involves learning subskills, and repeated practice.

Platform capabilities for skills and knowledge

This section describes the core capabilities for manipulating and managing skills and knowledge in Cloud Learning Platform, and how they relate back to the use cases identified above. It must be emphasized that solving many of the use cases identified above require applications that are integrated with CLP. CLP is designed to support educational applications by representing skills and knowledge and providing services to manage and curate skills and

knowledge, and map skills and knowledge to learning content. Users will employ these applications to solve their use cases.

We are designing and developing the services for skills and knowledge in Cloud Learning Platform jointly as we develop the initial applications that use those capabilities. Thus we expect these capabilities to be refined as the applications themselves are developed and tested in real-world deployments.

Knowledge

- **representing knowledge in tree graphs**, with a hierarchical structure.
 - Entities are nodes and both compositional relations and general relations are supported
- **importing knowledge from content** sources
 - extracting that knowledge from text and images into knowledge graphs
- **topic modeling on content** to infer a tree structure for the knowledge graph

Skills

- **representing skills as entities in tree graphs** with a hierarchical structure
 - skills are entities. compositional (parent/child) relations are supported.
 - directed edges in the graph can be stored to represent precedent relations
 - skill objects follow a data schema that is general enough to represent all the skill data in common repositories used in education
 - skills attributes include a text description and other attributes like links to the data source of the skill
- **importing skills** as entities from external repositories of skills
 - Currently support CASE, OSN, CTDL
- **mapping skills to the knowledge** (stored in knowledge graphs in the platform) that represent the context of the skill
- **comparing skills**, by comparing both the attributes of the skill as well as the mapping of the skill.
- **associating assessment** (represented in data models for those assessments) with the skill object model

Assessment

- **generating formative assessment** from skill entities (including all the associated meta data) and the knowledge context for those skills.

Learning Goals and Learning Paths

The platform has the ability to represent the learning goals of a student mapped to a set of skills and knowledge. The platform also can represent the pathways (sequences of learning experiences) available to acquire skills and knowledge. We will document these capabilities in detail elsewhere, but given the foundational nature of skills and knowledge to the use cases associated with learning goals and the respective platform components, it is worth mentioning here.

Examples

Skill examples

Here are some examples of Skills, and associated subskills, and knowledge components:

Skill (Competency): Manage drug treatment for a patient

Sub-Skill: Manage the drug treatment regimen for a patient with a chronic disease

Knowledge component(s): Chronic diseases

Knowledge component(s): Drugs used to treat chronic diseases

Knowledge component(s): Drug treatment regimens for chronic diseases

Skill (SubCompetency): Evaluate drug regimen compliance for a patient

Sub-Skill: Evaluate drug compliance for diabetes type 2 patients

Knowledge component: Drugs for Diabetes type 2

Knowledge component: Drug treatment regimens for Diabetes type 2

Learning units

The leaf nodes of the KG tree are termed “Learning Units” (LUs), and represent independent, granular items of knowledge at the lowest levels of Blooms taxonomy, the “recall” skills. So there is technically a recall skill associated with each Learning Unit. In practice we do not represent the recall skills in the Skill tree, since there is a simple 1-1 relationship between an LU and a recall Skill.

Each LU is associated with a Knowledge Graph that encodes entities and relations extracted from learning content associated with that LU. LU's are also associated with a set of text extracts from learning content (the “learning context”), that provide context for that LU. For example:

Learning Unit: Nateglinide is a Meglitinides drug used to treat diabetes type 2.

Skill (recall): Recall drugs used to treat diabetes type 2 in the Meglitinides class.

Knowledge Component: Nateglinide is a Meglitinides drug used to treat diabetes type 2.

Learning context:

<sections of text from a nursing textbook on the treatment of diabetes type 2>

References

[W3C Semantic Ontologies](#)

[OSN- Technical Solutions for a Skills-based Future](#)

Note: “Glidepath” is the project name used in our SNHU project for Cloud Learning Platform

- Cloud Learning Platform [Skill Service PRD](#)
- Cloud Learning Platform [Skill Service TDD](#)